

Applications of Radian Units for the Measure of Angles

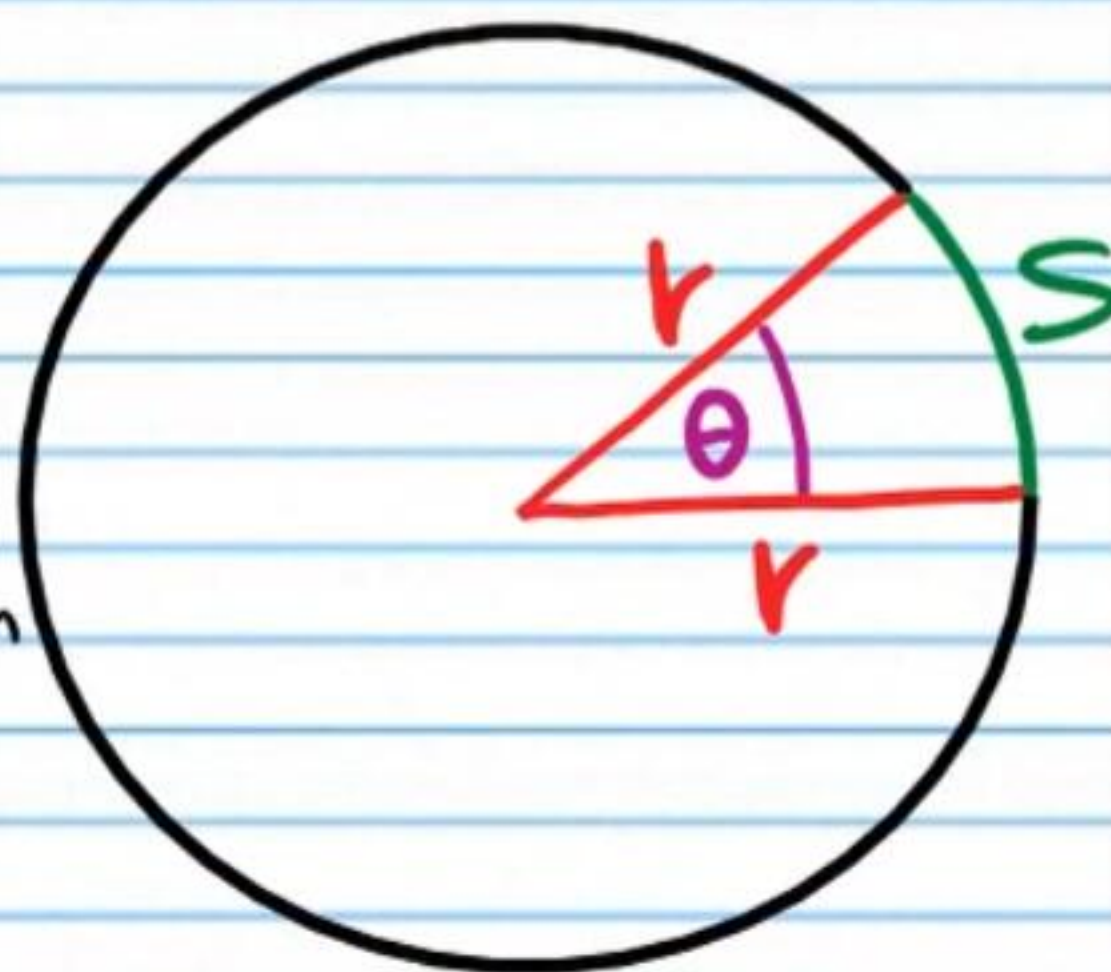
Arc Length

If θ is measured in degree units, then

$$s = \frac{\theta}{360^\circ} 2\pi r$$

If θ is measured in radian units, then

$$s = \frac{\theta}{2\pi} 2\pi r$$



$$s = \theta r$$

* This formula can be applied with radian units only.

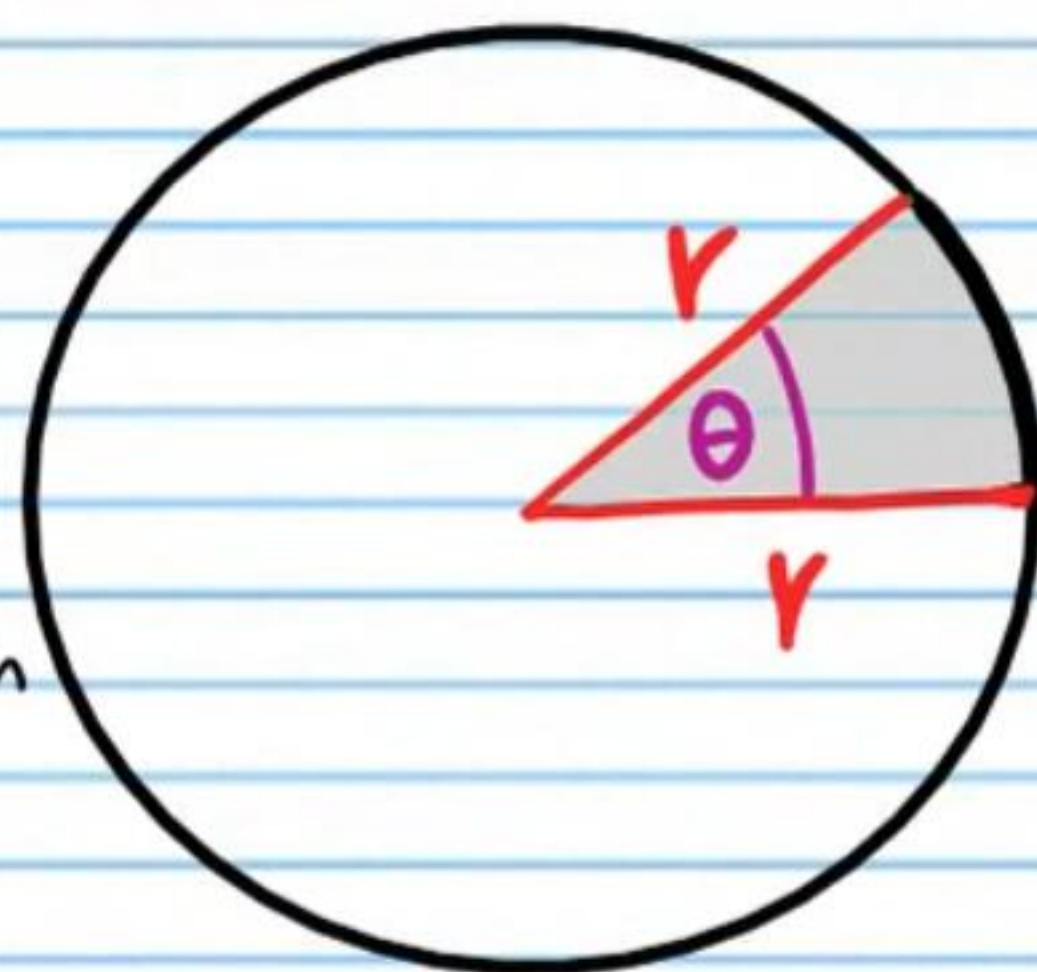
The Area of a Sector

If θ is measured in degree units, then

$$A = \frac{\theta}{360^\circ} \pi r^2$$

If θ is measured in radian units, then

$$A = \frac{\theta}{2\pi} \pi r^2$$



$$A = \frac{\theta}{2} r^2$$

* This formula can be applied with radian units only.

Practice

- ① Find the length of an arc on a circle with radius 3 centimeters and central angle $\frac{\pi}{7}$.

$$s = r\theta = (3)\left(\frac{\pi}{7}\right) = \boxed{\frac{3\pi}{7} \text{ cm}}$$

- ② Find the length of an arc on a circle with radius 4.5 feet and central angle $\frac{8\pi}{5}$.

$$s = r\theta = (4.5)\left(\frac{8\pi}{5}\right) = \boxed{\frac{36\pi}{5} \text{ ft}}$$

- ③ Find the length of an arc on a circle with radius 2 meters and central angle $\frac{16\pi}{9}$.

$$s = r\theta = (2)\left(\frac{16\pi}{9}\right) = \boxed{\frac{32\pi}{9} \text{ m}}$$

- ④ Find the length of an arc on a circle with radius 8 inches and central angle $\frac{\pi}{8}$.

$$s = r\theta = (8)\left(\frac{\pi}{8}\right) = \boxed{\pi \text{ in}}$$

- ⑤ Find the area of a sector of a circle with radius 20 millimeters and central angle $\frac{\pi}{5}$. Round to the nearest tenth.

$$A = \frac{\theta}{2} r^2 = \frac{\pi}{5} (20)^2 \approx \boxed{125.7 \text{ mm}^2}$$

⑥ Find the area of a sector of a circle with radius 6.23 miles and central angle $\frac{6\pi}{7}$. Round to the nearest thousandth.

$$A = \frac{\theta}{2} r^2 = \frac{\frac{6\pi}{7}}{2} (6.23)^2 \approx \boxed{52.258 \text{ mi}^2}$$

⑦ Find the area of a sector of a circle with radius 32 feet and central angle $\frac{5\pi}{9}$. Round to the nearest hundredth.

$$A = \frac{\theta}{2} r^2 = \frac{\frac{5\pi}{9}}{2} (32)^2 \approx \boxed{893.61 \text{ ft}^2}$$

⑧ Find the area of a sector of a circle with radius 2 kilometers and central angle $\frac{13\pi}{8}$. Round to the nearest whole number.

$$A = \frac{\theta}{2} r^2 = \frac{\frac{13\pi}{8}}{2} (2)^2 \approx \boxed{10 \text{ km}^2}$$

If an object moves s units on a circular path in t units of time, then V is defined as the **linear velocity** according to the equation below.

$$V = \frac{s}{t}$$

If an object on a circle rotates θ angle units in t units of time, then ω is defined as the **angular velocity** according to the equation below.

$$\omega = \frac{\theta}{t}$$

Because $s = r\theta$ when θ is measured in radians, and $V = \frac{s}{t}$, then

$$V = \frac{s}{t} = \frac{r\theta}{t}$$

Furthermore, because $\omega = \frac{\theta}{t}$, the equation above can be modified to

$$V = r \frac{\theta}{t}$$

$$V = r\omega$$

$$\boxed{V = r\omega}$$

★ This formula can be applied with radian units only.

Practice

- ⑨ An object moves on a circular path with an angular velocity of $\frac{3\pi}{7} \text{ rad/s}$. What is the object's linear velocity if the radius of the path is 2 inches?

$$V = r\omega = (2)\left(\frac{3\pi}{7}\right) = \boxed{\frac{6\pi}{7} \text{ in/s}}$$

- ⑩ An object moves on a circular path with a linear velocity of 8 m/min . What is the object's angular velocity if the radius of the path is 22 meters?

$$V = r\omega$$

$$8 = 22\omega$$

$$\omega = \frac{4}{11} \text{ rad/min}$$

- ⑪ An object moves on a circular path with an angular velocity of $\frac{15\pi}{8} \text{ rad/h}$. What is the object's linear velocity if the radius of the path is 12 feet?

Hours.

$$V = r\omega = (12)\left(\frac{15\pi}{8}\right) = \boxed{\frac{45\pi}{2} \text{ ft/h}}$$

- ⑫ An object moves on a circular path with a linear velocity of 9 cm/day . What is the object's angular velocity if the radius of the path is 3 cm?

$$V = r\omega$$

$$9 = 3\omega$$

$$\omega = 3 \text{ rad/day}$$

Converting Rates

Revolution: A complete rotation.

Revolutions per minute = rpm

1 Revolution = 2π radians = 360°

- ⑬ The maximum rpm of a particular motorcycle engine is 13,000. Convert this rate to radians per second. Round to the nearest whole number.

$$\omega = \frac{13,000 \text{ rev}}{1 \text{ min}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} \times \frac{1 \text{ min}}{60 \text{ s}} = \frac{26,000\pi \text{ rad}}{60 \text{ s}}$$

$$\approx \boxed{1,361 \text{ rad/s}}$$

- ⑭ A bicycle wheel rotates $\frac{10\pi}{7} \text{ rad/s}$. Convert this rate to rpm. Round to the nearest hundredth.

$$\frac{\frac{10\pi}{7} \text{ rad}}{1 \text{ s}} \times \frac{1 \text{ rev}}{2\pi \text{ rad}} \times \frac{60 \text{ s}}{1 \text{ min}} = \frac{600\pi \text{ rev}}{7 \cdot 2\pi \text{ min}}$$

$$\approx 42.86 \text{ rev/min}$$

$$\approx \boxed{42.86 \text{ rpm}}$$

- ⑮ A Ferris wheel rotates $2,447^\circ/\text{h}$. Convert this rate to rpm.
Round to the nearest tenth.

$$\frac{2,447^\circ}{1 \text{ h}} \times \frac{1 \text{ rev}}{360^\circ} \times \frac{1 \text{ h}}{60 \text{ min}} = \frac{2,447 \text{ rev}}{21,600 \text{ min}} \approx \boxed{0.1 \text{ rpm}}$$

- ⑯ A saw blade rotates 1,200 rpm. Convert this rate to radians per second. Round to the nearest tenth.

$$\frac{1,200 \text{ rev}}{1 \text{ min}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} \times \frac{1 \text{ min}}{60 \text{ s}} = \frac{2400\pi \text{ rad}}{60 \text{ s}} \approx \boxed{125.7 \text{ rad/s}}$$

- ⑰ A hard drive rotates $82\pi/7 \text{ rad/s}$. Convert this rate to rpm.
Round to the nearest thousandth.

$$\frac{82\pi \text{ rad}}{7 \text{ s}} \times \frac{1 \text{ rev}}{2\pi \text{ rad}} \times \frac{60 \text{ s}}{1 \text{ min}} = \frac{4,920\pi \text{ rev}}{7 \times 2\pi \text{ min}} \approx \boxed{351.429 \text{ rpm}}$$

- ⑱ A windmill rotates $65,000^\circ/\text{h}$. Convert this rate to rpm.
Round to the nearest whole number.

$$\frac{65,000^\circ}{1 \text{ h}} \times \frac{1 \text{ rev}}{360^\circ} \times \frac{1 \text{ h}}{60 \text{ min}} = \frac{65,000 \text{ rev}}{21,600 \text{ min}} \approx \boxed{3 \text{ rpm}}$$

Challenge Problems

- ⑲ The wheel of a car rotates at 208 rpm. How fast does the outer edge of the tire rotate if its radius is 1.2 feet? Round to the nearest hundredth.

$$W = 208 \text{ rev/min}, r = 1.2 \text{ ft}, V = ?$$

$$\text{I) } \frac{208 \text{ rev}}{1 \text{ min}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} = \frac{416\pi \text{ rad}}{1 \text{ min}} = 416\pi \text{ rad/min}$$

$$\text{II) } V = rW = (1.2)(416\pi) \approx \boxed{1,568.28 \text{ ft/min}}$$

- ⑳ The earth completes one rotation in 24 hours. If the diameter of the earth is 8,000 miles, how fast does a person at the equator move relative to the center of the earth? Round to the nearest whole number.

$$W = \frac{1 \text{ rotation}}{24 \text{ h}} = \frac{1 \text{ rev}}{24 \text{ h}}, r = \frac{D}{2} = \frac{8,000}{2} = 4,000 \text{ mi}$$

$$\text{I) } \frac{1 \text{ rev}}{24 \text{ h}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} = \frac{2\pi \text{ rad}}{24 \text{ h}} = \frac{\pi}{12} \text{ rad/h}$$

$$\text{II) } V = rW = (4,000)\left(\frac{\pi}{12}\right) \approx \boxed{1,047 \text{ mi/h}}$$

- (21) A man swings a rock tied to a rope on a circular path at 136 rpm. If the length of the rope from his hand to the rock is 30 inches, how fast does the rock move? Round to the nearest whole number.

$$\omega = \frac{136 \text{ rev}}{1 \text{ min}}, \quad r = 30 \text{ in}$$

$$\text{I) } \frac{136 \text{ rev}}{1 \text{ min}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} = \frac{272\pi \text{ rad}}{1 \text{ min}} = 272\pi \text{ rad/min}$$

$$\text{II) } v = r\omega = (30)(272\pi) \approx \boxed{25,635 \text{ in/min}}$$

- (22) A man spins a basketball on his finger with a constant angular velocity. The ball completes one rotation in 0.12 seconds and the diameter of the ball is 9.5 inches. If the rotation of the ball is perpendicular to the man's finger, find the linear velocity of the fastest moving part of the ball. Round to the nearest tenth.

$$\omega = \frac{1 \text{ rotation}}{0.12 \text{ s}} = \frac{1 \text{ rev}}{0.12 \text{ s}}, \quad r = \frac{D}{2} = \frac{9.5}{2} = 4.75 \text{ in}$$

$$\text{I) } \omega = \frac{1 \text{ rev}}{0.12 \text{ s}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} = \frac{2\pi \text{ rad}}{0.12 \text{ s}}$$

$$\text{II) } v = r\omega = (4.75)\left(\frac{2\pi}{0.12}\right) \approx \boxed{248.7 \text{ in/s}}$$

- (23) A toy train moves on a circular track. It completes 5 laps in 1.5 minutes. If the radius of the track is 100 inches, how fast does the train move in inches per second? Round to the nearest hundredth.

$$\omega = \frac{5 \text{ laps}}{1.5 \text{ min}} = \frac{5 \text{ rev}}{1.5 \text{ min}}, \quad r = 100 \text{ in}$$

$$\text{I) } \omega = \frac{5 \text{ rev}}{1.5 \text{ min}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} = \frac{10\pi \text{ rad}}{1.5 \text{ min}}$$

$$\text{II) } v = r\omega = (100)\left(\frac{10\pi}{1.5}\right) \approx \frac{1000\pi \text{ in}}{1.5 \text{ min}}$$

$$\text{III) } v = \frac{1000\pi \text{ in}}{1.5 \text{ min}} \times \frac{1 \text{ min}}{60 \text{ s}} = \frac{1000\pi \text{ in}}{90 \text{ s}}$$

$$\approx \boxed{34.91 \text{ in/s}}$$

- (24) Linsey runs on a circular track at a constant pace. She runs 3 laps in 16 minutes. If the radius of the track is 38 meters, how many meters does she run per second? Round to the nearest thousandth.

$$\omega = \frac{3 \text{ laps}}{16 \text{ min}} = \frac{3 \text{ rev}}{16 \text{ min}}, \quad r = 38 \text{ m}$$

$$\text{I) } \omega = \frac{3 \text{ rev}}{16 \text{ min}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} = \frac{6\pi \text{ rad}}{16 \text{ min}} = \frac{3\pi \text{ rad}}{8 \text{ min}}$$

$$\text{II) } v = r\omega = (38)\left(\frac{3\pi}{8}\right) = \frac{114\pi \text{ m}}{8 \text{ min}}$$

$$\text{III) } v = \frac{114\pi \text{ m}}{8 \text{ min}} \times \frac{1 \text{ min}}{60 \text{ s}} = \frac{114\pi \text{ m}}{480 \text{ s}}$$

$$\approx \boxed{0.746 \text{ m/s}}$$

- (25) A pepperoni pizza sits on the center of a rotating oven. One pepperoni is 8 centimeters from the center of the pizza and moves 0.2 centimeters per second. What is the angular velocity of the oven? Another pepperoni is 10 centimeters from the center. How fast is this second pepperoni moving?

$$v_1 = 0.2 \text{ cm/s}, \quad r_1 = 8 \text{ cm}$$

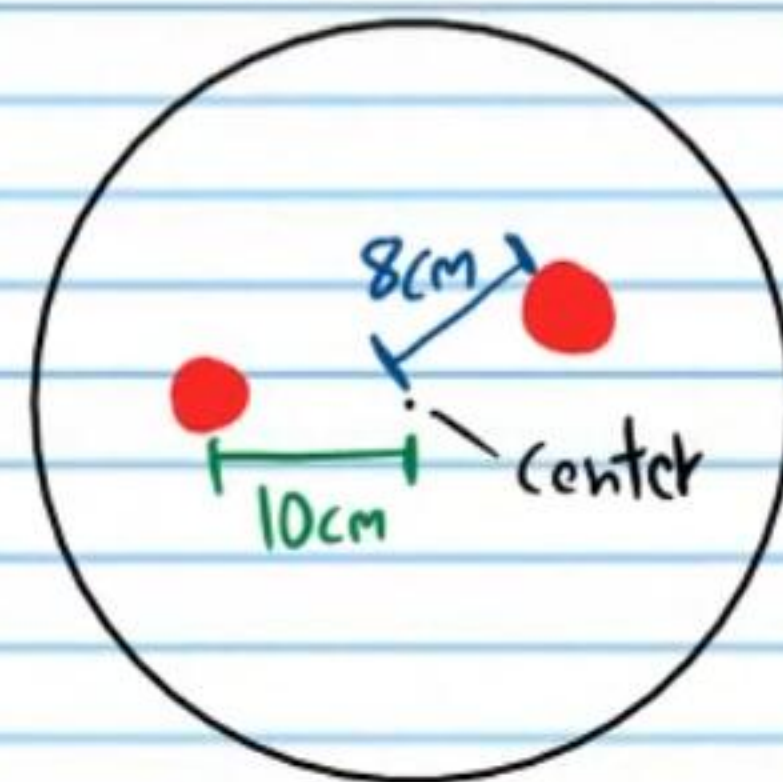
$$\text{I) } v_1 = r_1 \omega$$

$$0.2 = 8\omega$$

$$\boxed{\omega = 0.025 \text{ rad/s}}$$

$$v_2 = ? \text{ cm/s}, \quad r_2 = 10 \text{ cm}$$

$$\text{II) } v_2 = r_2 \omega = (10)(0.025) = \boxed{0.25 \text{ cm/s}}$$



②6 A boy pushes two other children on a merry-go-around at the park. One child is 6 feet from the axis of rotation and the other is 3 feet from the axis. If the child closer to the center is moving 8 feet per second, what is the angular velocity of the merry-go-around? What is the linear velocity of the other child?

$$V_1 = 8 \text{ ft/s}, r_1 = 3 \text{ ft}$$

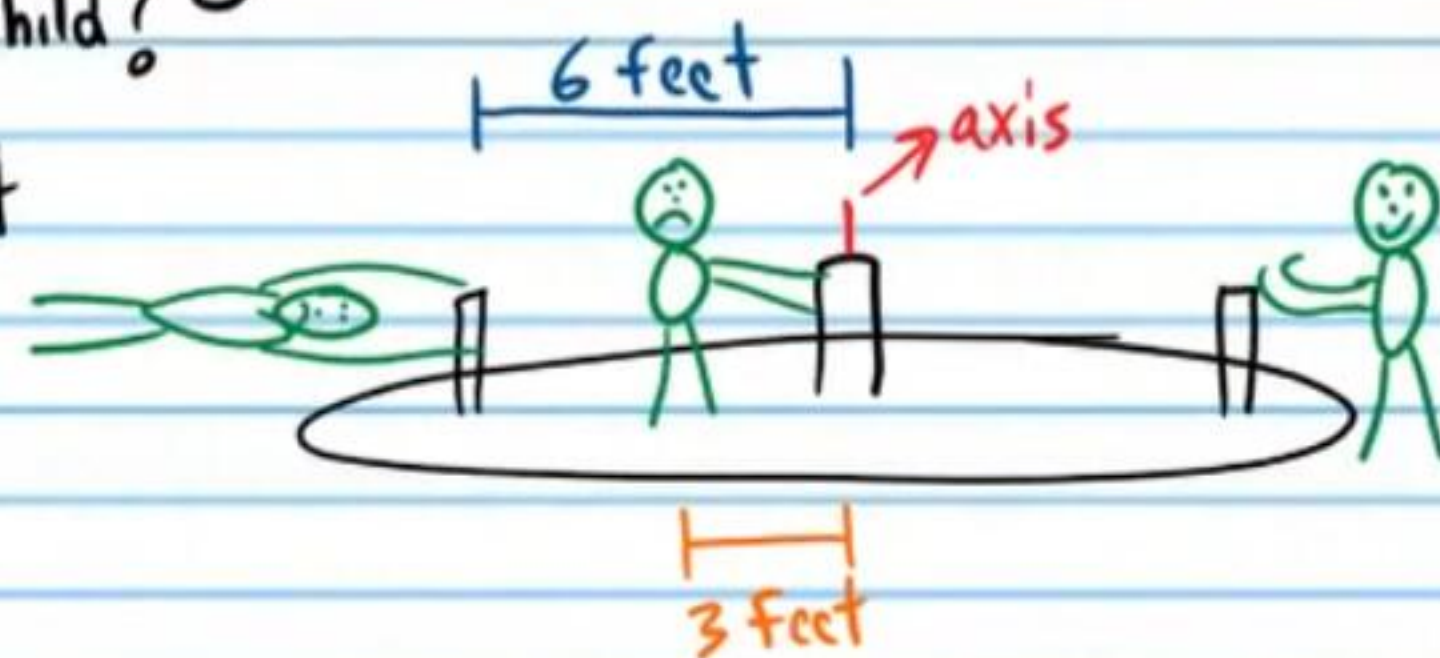
$$\text{I) } V_1 = r_1 \omega$$

$$8 = 3\omega$$

$$\boxed{\omega = \frac{8}{3} \text{ rad/s}}$$

$$V_2 = ? \text{ ft/s}, r_2 = 6 \text{ ft}$$

$$V_2 = r_2 \omega = (6) \left(\frac{8}{3} \right) = \boxed{16 \text{ ft/s}}$$



• ②7 A bit of paint on a sprocket 15 millimeters from its center moves 3 millimeters per second. What is the angular velocity of the sprocket? How fast does a bit of paint 26 millimeters from the center move?

$$V_1 = 3 \text{ mm/s}, r_1 = 15 \text{ mm}$$

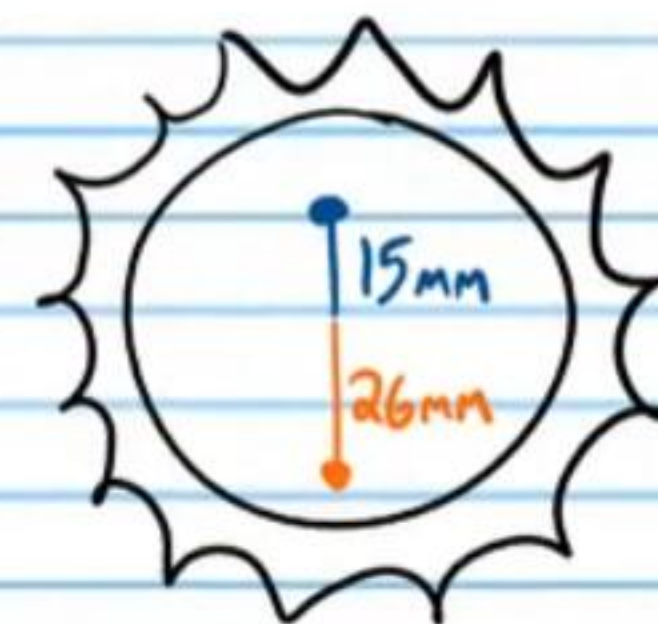
$$\text{I) } V_1 = r_1 \omega$$

$$3 = 15\omega$$

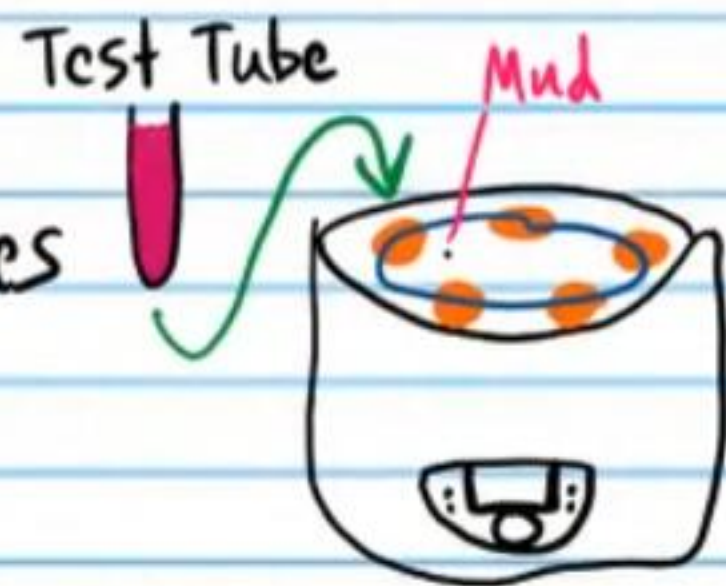
$$\boxed{\omega = \frac{1}{5} \text{ rad/s}}$$

$$V_2 = ? \text{ mm/s}, r_2 = 26 \text{ mm}$$

$$\text{II) } V_2 = r_2 \omega = (26) \left(\frac{1}{5} \right) = \boxed{5.2 \text{ mm/s}}$$



28) Contents in a liquid mixture are often separated by spinning the mixture in a centrifuge. A test tube is placed 4 inches from the center of the device and when its maximum speed is reached, the tube moves at 1,800 inches per second. Find the angular velocity of the centrifuge and the linear velocity of the speck of mud that is 3.4 inches from the center of the device.



$$V_1 = 1,800 \text{ in/s}, r_1 = 4 \text{ in}$$

$$\text{I) } V_1 = r_1 \omega$$

$$1800 = 4\omega$$

$$\boxed{\omega = 450 \text{ rad/s}}$$

$$V_2 = ? \text{ in/s}, r_2 = 3.4 \text{ in}$$

$$\text{II) } V_2 = r_2 \omega = (3.4)(450) = \boxed{1,530 \text{ in/s}}$$